

Delta 0.5 diagnostic summary

Case. $r = 0.02$, $\delta_1 = \delta_2 = 0.5$, $T = 1$, $N_{\text{ex}} = 12$, $v_0 = 0.114$, $v'_0 = 0.110$. All outputs are under D:/Mara PhD/Paper-C/To be deleted.

Conclusion. The diagnostics do not support a stale benchmark or wrong-parameter explanation. The sandbox value for $K=100$, $M=48$, $N=60,000$ is 11.508319355383, and the manuscript CSV value is 11.508319355383; the absolute difference is 0.000000000000. For $K=100$, plain LSMC changes by 1.571 between $M=48$ and $M=1200$ at $N=60,000$, while increasing paths at $M=48$ from 60,000 to 600,000 changes the price by 0.124. The targeted Figure 5 benchmark check at $M=100$ and $N=500,000$ gives $K=100$ price 10.914; across strikes its relative errors remain between 8.16

LSMC step and path convergence. Low denotes the independent validation estimator reported by the copied LSMC engine; Gap is $|\text{direct} - \text{low}|/|\text{direct}|$.

Case/M	N	Direct	SE	Low	Gap
48	60000	11.508	0.080	11.424	0.0073
96	60000	10.956	0.076	10.967	0.0010
240	60000	10.440	0.075	10.530	0.0087
600	60000	10.048	0.074	10.289	0.0240
1200	60000	9.937	0.074	10.005	0.0068
48	600000	11.384	0.025	11.444	0.0053
96	600000	10.944	0.024	10.936	0.0007
240	600000	10.446	0.024	10.433	0.0013
600	600000	10.122	0.023	10.075	0.0047
1200	600000	9.946	0.023	9.906	0.0040
K=70, M=48	60000	4.051	0.046	4.022	0.0070
K=70, M=1200	60000	3.382	0.042	3.406	0.0071
K=110, M=48	60000	15.613	0.088	15.579	0.0022
K=110, M=1200	60000	13.783	0.079	13.976	0.0139

Targeted Figure 5 benchmark check. The benchmark values are the direct LSMC references used for Figure 5 of `main.springer.pdf`: $M_{\text{ref}} = 1200$ and $N_{\text{ref}} = 1,200,000$. The new run uses plain LSMC with $M = 100$ and $N = 500,000$.

K	Direct	SE	Figure 5 ref.	Rel. error	Low	Gap
70	3.775	0.015	3.364	12.22%	3.764	0.0028
80	5.529	0.019	4.961	11.45%	5.510	0.0035
90	7.857	0.023	7.082	10.93%	7.835	0.0027
100	10.914	0.026	9.921	10.01%	10.900	0.0013
110	14.987	0.029	13.857	8.16%	15.003	0.0010

Fresh M=500 benchmark check. This compares the targeted $M = 100$, $N = 500,000$ plain LSMC prices against new plain LSMC benchmarks generated in this sandbox with $M = 500$ and $N = 1,200,000$ for $K = 90$ and $K = 100$.

K	M=100 price	SE	M=500 bench.	SE	Diff.	Rel. diff.
90	7.857	0.023	7.276	0.014	0.580	7.98%
100	10.914	0.026	10.179	0.016	0.735	7.22%

All-path versus ITM-only policy.	K	M	All-path	ITM-only	Diff.	SE
	70	48	4.051	4.100	0.050	0.046
	70	1200	3.382	3.443	0.060	0.042
	100	48	11.508	11.534	0.026	0.080
	100	1200	9.937	9.961	0.024	0.074
	110	48	15.613	15.644	0.030	0.088
	110	1200	13.783	13.804	0.021	0.079

Forward Euler boundary diagnostic. This is a terminal European put diagnostic, so it does not use Bermudan regression.

M	neg v	zero v	neg v'	zero v'	Eur K=100	SE
24	20.9%	23.0%	5.1%	5.1%	11.789	0.030
48	18.3%	20.0%	4.2%	4.2%	11.203	0.030
96	16.2%	17.7%	3.5%	3.5%	10.783	0.029
240	14.0%	15.3%	3.0%	3.0%	10.258	0.029
600	12.4%	13.5%	2.5%	2.5%	9.931	0.028
1200	11.4%	12.3%	2.3%	2.3%	9.736	0.028

Hybrid-PDE sensitivity. For K=100 and N=20,000, Min grid and Max grid are the range across default, expanded, wide, and q995 grids. Other rows report the default grid.

Case	N	Default	SE	Min grid	Max grid	Range
K=100, M=48	20000	11.226	0.042	11.226	11.270	0.044
K=100, M=96	20000	10.887	0.042	10.882	10.923	0.041
K=100, M=240	20000	10.487	0.042	10.487	10.521	0.033
K=70, M=48	20000	3.871	0.017	-	-	-
K=70, M=96	20000	3.720	0.017	-	-	-
K=110, M=48	20000	15.529	0.053	-	-	-
K=110, M=96	20000	15.141	0.053	-	-	-
K=100, M=48	60000	11.371	0.024	-	-	-
K=100, M=96	60000	10.861	0.024	-	-	-

Recommendation. Do not treat the large relative errors as a path-count failure or a benchmark-file mismatch. If this parameter case remains in the manuscript, it should be framed as a stress test showing sensitivity of the square-root case to the Euler time step, or rerun with a finer-step design and a discretization scheme better suited to the boundary.